SECTION 194
TRAFFIC MANAGEMENT FOR WORK ZONES

PART A – GENERAL

194.01 General – Traffic management requirements shall be as specified herein and as specified in the Special Provisions or by the Ministry Representative.

The Contractor's Traffic Management Plan, Construction Schedule and its provisions for traffic control and safety shall be based on providing safe passage through the Work Zone for all road users (i.e. cyclists, pedestrians and motorists). The Contractor shall at all times make provisions for traffic through the Site to a sufficiently high standard to ensure the safety of the workers on the Project, the safety and the convenience of the travelling public, driver expectancy, and the protection of the Work Area. The Contractor shall supply and maintain all traffic control devices and personnel necessary to provide all traffic control.

The Contractor's accommodation for pedestrian, bicycle and vehicular traffic, and proposed traffic control procedures for each work operation will require acceptance by the Ministry Representative prior to the start of that operation.

Any one or more of the advance warning area, transition area, buffer space and termination area of the work zone may be outside the Limits of Construction, but this will in no way diminish the Contractor's responsibility to meet the requirements of the Traffic Management Plan.

194.02 Definitions – The following definition applies within this Section:

(a) **"Traffic Management Manual" or "TMM"** means the Ministry's *2020 Traffic Management Manual for Work on Roadways*, as it may be amended or replaced from time to time.

The TMM is available on-line at:

https://www2.gov.bc.ca/gov/content/transportation/transportation-infrastructure/engineering-standards-guidelines/traffic-engineering-safety/trafficmanagementmanual

(b) **Other general terms** used within this Section are defined in TMM Appendix A.1, including without limitation,:

(i) Construction speed zone

(ii) Traffic Management Plan

(iii) Traffic Control Plan

(iv) Incident Management Plan

(v) Public Information Plan

(vi) Implementation Plan

(vii) Work Activity Area

(viii) Work Zone

194.03 Standard Documents – Traffic strategies and plans shall be developed in accordance with standards defined in the latest version of the following documents:

(a) The *Traffic Management Manual*

(b) The Ministry's *Electrical and Traffic Engineering Manual*

(c) All current Ministry Technical Circulars and Technical Bulletins

(d) *TAC Geometric Design Guide for Canadian Roads*

(e) The Ministry's *B.C. Supplement to TAC Geometric Design Guide*

(f) Workers' Compensation Board (WCB) *Occupational Health and Safety Regulation*

All Ministry publications listed above may be found, free of charge, on-line through:

https://www2.gov.bc.ca/gov/content/transportation/transportation-infrastructure/engineering-standards-guidelines

or can be purchased from:

CROWN PUBLICATIONS
(King's Printer for British Columbia)
PO BOX 9452 STN PROV GOVT
VICTORIA B.C. V8W 9V7
https://www.crownpub.bc.ca/Home/About
Email: crownpub@gov.bc.ca
PHONE: 1-800-663-6105
FAX: (250) 387-1120

194.04 Traffic Control Supervisor – The Contractor shall designate a Traffic Control Supervisor who has the Contractor's authority to respond to traffic control requirements and who shall provide surveillance of traffic conditions through the Work Zone. This person, or qualified designate, shall be available at all times, both when the Contractor is working and during weekends or other periods when the Contractor's forces are not active on the Project, to respond to calls from the police and from representatives of the Ministry concerning damage or deficiencies in the traffic control devices and the passage for traffic through the Work Zone.

The traffic control supervisor shall meet the requirements of the TMM and WCB Regulation and shall not be the Contractor's site superintendent.

194.05 Compliance with Acts and Regulations – The Contractor shall comply with the requirements of any legislation, Regulations and by-laws as are in force for the regulation of traffic or use of any road upon or over which it is necessary to carry out work or to haul materials or things.

PART B – PLANNING AND SCHEDULING

194.11 Provide Traffic Management Plan – The Contractor shall prepare a Traffic Management Plan (TMP) in accordance with the Ministry definitions and guidelines specified in the Traffic Management Manual.

The Special Provisions or the Ministry Representative will specify any required traffic management conditions, strategies, and required documentation for the Traffic Management Plan.

The Traffic Management Plan shall be fully integrated with the Contractor's plan and schedule for carrying out the work, comply with the requirements of the TMM, the conditions and strategies listed below, and with the provisions of the Contract.

194.12 Traffic Management Plan Review – The Ministry Representative may review the Contractor's Traffic Management Plan for general conformance with the requirements of the Special Provisions and the Contract for sufficiency of detail to provide a complete and unambiguous traffic control layout. No review or change requested by the Ministry Representative in any way relieves the Contractor of any of its responsibilities for ensuring safe and adequate traffic management. No work that affects traffic will be permitted until the Ministry Representative accepts the Traffic Management Plan.

Where the TMM shows a choice of traffic control layouts for a given specific situation, the Ministry Representative may request the Contractor to use an alternative layout which is acceptable.

194.12.01 Review Time Periods – The following time periods, subject to SS 145.34, are required for plan review and acceptance unless otherwise specified in the Special Provisions or by the Ministry Representative:

(a) A Traffic Control Plan shall be submitted to the Ministry Representative for review at least fourteen (14) days prior to the start of any Works effecting traffic flows. The following conditions shall apply:

- **(i)** Any plan modifications shall be submitted to the Ministry Representative for review at least fourteen (14) days prior to any changes being made in the field.
- **(ii)** A staged plan shall be submitted to the Ministry Representative for review at least fourteen (14) days prior to implementation of a new stage.

(b) The Implementation Plan, if required, shall be submitted to the Ministry Representative for review at least twenty-one (21) days prior to the start of any Works effecting traffic flows.

(c) The Public Information Plan, if required, shall be submitted to the Ministry Representative for review at least twenty-one (21) days prior to the start of any Works effecting traffic flows.

(d) The Incident Management Plan, if required, shall be submitted to the Ministry Representative for review at least twenty-one (21) days prior to the start of any Works effecting traffic flows.

PART C – TRAFFIC MANAGEMENT OPERATIONAL DETAILS

194.21 Speed Zones – Reduction in speed limits may take several forms as defined in the TMM 2.4 Management of Speed. Construction speed zones within the Project will be determined at the discretion of the Road Authority or as specified in the Special Provisions.

The Contractor's work plan, Construction Schedule and its provisions for traffic control and safety shall include references to any reduced posted speed limits.

The location and length of posted construction speed zones shall be evaluated daily, or more frequently as appropriate to the Work, to provide the shortest construction speed zone practical. For example, on a resurfacing project, the reduced speed limit would usually be invoked around active operations such as milling, priming, paving, and, if necessary, at access points to pits and the plant, but will not generally be acceptable for areas of the Site used only for haul.

194.22 Use of Channelizing Devices and Taper Rates – Traffic cones, tubular markers, barrels/drums, barricades or other channelizing devices shall be used in the following circumstances for guiding and directing road users:

(a) from one lane to another or for merging traffic because of a reduction in the number of lanes

(b) around a hazard on the travelled way

(c) through an area of unfinished construction where the intended traffic lanes are not self-evident

(d) at any other location where, in the Ministry Representative's opinion, channelizing will ensure the safe passage of public traffic.

Minimum taper rates and spacing between devices shall be as shown in TMM.

194.23 Equipment on the Travelled Way – When working on the travelled way, equipment shall be positioned to present a minimum of interference and hazard to the travelling public. Equipment units shall be kept as close together as working conditions will permit and preferably on the same side of the travelled way. Equipment shall not

be left overnight on the travelled way but shall be located so as to avoid the possibility of collision.

194.24 Roadway Configuration – The Contractor shall arrange to keep interconnected lanes of the highway open to traffic at all times with the least amount of inconvenience to the travelling public unless otherwise authorized by the Ministry Representative.

194.25 Roadway Condition – The Contractor shall keep the travelled way well-graded, free of potholes and of sufficient width that the required number of lanes of traffic may pass. The subgrade shall be completed progressively in order that gravel surfacing may follow in an orderly manner. Prior to tearing up a portion of the existing travelled way the Contractor shall obtain Ministry Representative approval and shall also have available a sufficient supply of crushed gravel, or another approved product, to ensure a smooth riding surface during the work.

194.26 Closures – Road closures shall be in accordance with the Special Provisions.

Road closures will not be permitted until the Contractor requests and receives written permission from the Ministry Representative. This request shall be submitted in writing not less than fourteen (14) days in advance of the closure.

Upon receiving written permission for the closure, the Contractor must notify the local police, fire, ambulance, municipality, school board, and public transit at least seven (7) days in advance of the closure.

The public must be notified by local radio and newspaper at least seven (7) days in advance of the closure. The Contractor shall arrange any public announcement relating to the Contract and incorporate the Ministry logo through the Ministry Representative and Corporate Writing Services. Such advertisements will be made at the Ministry's expense. Notices or other advertisements of the Contractor unrelated to the Contract, or where the Ministry's logo or name is not incorporated, may be made by the Contractor at the Contractor's expense.

194.27 Detours – Where detours or diversions are required for the execution of the work or deemed necessary by the Ministry Representative, the Contractor shall provide detours or temporary roads to facilitate the passage of traffic around the restricted construction area in accordance with Special Provisions or as otherwise directed by the Ministry Representative. The Contractor shall be responsible for all signs and their maintenance, and maintenance of the detour or temporary road. The Contractor shall consider the condition of the pavement used for detours and its impact on the safety and function of the detour. Milled surfaces upon which traffic is to run shall be clean and allow adequate drainage.

All detours shall be adequately signed such that the detour route is clear and unambiguous to all road users.

The minimum design requirements for any detours will be specified in the Special Provisions or by the Ministry Representative.

Where the anticipated detour or diversion extends beyond the existing road surface, the Contractor shall submit an engineering design to the Ministry Representative for review, a minimum of fourteen (14) days in advance of building the detour. The engineering design shall include the design speed, the horizontal and vertical geometry, plus evidence that the design vehicle is accommodated for all movements and that all the requirements of this Contract are satisfied.

Where the anticipated deviation is minor, the Contractor may apply to the Ministry Representative, in writing, for approval to proceed without a formal design. Such approval does not relieve the Contractor of any of its responsibilities for provision of a safe site.

194.28 Access to Property – During operations the Contractor shall provide and maintain reasonable road access and egress to property fronting along, or in the vicinity of, the work unless other reasonable means of road access exists. The Ministry Representative will be the sole judge of what may be deemed reasonable road access.

194.29 Seasonal Lay-Up – In the event that the Work is stopped due to seasonal lay-up conditions (typically winter conditions or high traffic periods), the Site is to be left in a condition as specified in the Special Provisions or as deemed acceptable by the Ministry Representative.

194.30 Delays – All acceptable major and minor delays to the travelling public will be determined by the Ministry Representative.

The Contractor shall monitor the queue lengths created by any delay to the flow of traffic. Traffic must be cleared prior to starting another delay unless authorized by the Ministry Representative. If the Ministry Representative determines that the traffic delays are excessive, the Contractor shall immediately cease construction activities and make all the travel lanes available to traffic as quickly as possible. Resumption of roadway operations shall be permitted as traffic levels dictate and upon approval by the Ministry Representative.

PART D – TRAFFIC CONTROL DEVICES

194.41 Portable Traffic Signals – Portable traffic signals, in accordance with TMM 4.8, may be used to provide bi-directional traffic control. Portable traffic signals are used primarily on longer term work zones. Typically, a pair of signals is set up at the perimeter of a roadway construction site.

Each signal unit shall have at least two signal heads for each approach and shall be positioned so that at least one signal

head is overhead and one is side-mounted. Each signal head shall have a 3 lamp (red, yellow, green) signal display.

194.41.01 Portable Traffic Signal Categories – The use of portable traffic signals falls into two categories:

(a) **Category 1** is a fixed time signal used for short duration work in low speed environments of ≤60 km/h, and where advance warning flashers are not required. A traffic engineer is not required to prepare the timing sheet for the signal.

(b) **Category 2** is an actuated or fixed time signal used for long duration work, and/or in high speed environments ≥70 km/h, and/or where advance warning flashers are required. A Traffic Engineer is required to prepare the timing sheet for the signal.

194.41.02 Use of Portable Traffic Signals

(a) **Request for Use –** The Contractor shall submit a request for the use of portable traffic signals to the Ministry Representative that includes:

(i) The site layout

(ii) Anticipated volumes

(iii) Requirements for advance warning signs

(iv) The category of portable traffic signal proposed

(v) Proposed signal timing plan(s)

(b) **Acceptance for Use –** The acceptance of the signal installation will be determined by the Ministry Representative.

If the proposed traffic signal installation is accepted by the Ministry Representative, the Contractor shall:

(i) Implement any portable traffic signals in accordance with the Traffic Management Manual

(ii) For category 2 signals, prepare a traffic signal timing plan, signed and sealed by a Traffic Engineer registered with the APEGBC

(iii) Supply and install all traffic controller equipment

(iv) Supply power

(v) Operate and maintain the signals

(c) **Technology –** Traffic controller technology used shall comply with industry standard practice and shall have:

(i) Manual override to hold the signal in green

(ii) A conflict monitoring mechanism that will ensure both signals in a pair cannot show green simultaneously

(iii) Ability to revert to flashing red mode if a fault is detected

(d) **Impact to Existing Signal –** Where the work activity impacts the operation of an existing signal, the Contractor must submit a written notification to the Ministry Representative a minimum of ten (10) days in advance of the disruption to the signal.

194.42 Vehicles as Traffic Control Devices – The use of shadow vehicles, buffer vehicles, vehicle-mounted crash attenuators, and/or pilot cars may be specified or warranted, and shall be supplied and operated in accordance with the TMM.

When required, the Contractor shall provide sufficient pilot cars, as required, to keep a steady and controlled flow of traffic moving around or through the construction area.

194.43 Use of Traffic Control Persons (TCP) – It is generally expected that TCP and/or Portable Traffic Signals will be required in the following situations:

(a) when public traffic is required to pass working vehicles or equipment which may block all or part of the travelled roadway.

(b) when it is necessary to institute a one-way traffic system through a construction area or other blockage where traffic volumes are heavy, approach speeds are high, and a traffic signal system is not in use.

(c) in high volume areas where temporary protection is required while other traffic control devices (barricades, cones, signs, etc.) are being erected or taken down.

(d) for emergency protection when other traffic control devices are not readily available.

(e) in all situations, where adequate protection for workers, working equipment and public traffic is not provided by other traffic control devices.

(f) Initiate traffic stoppages and reinstate flow of traffic to facilitate construction activities.

194.44 Traffic Signs – All standard signs, new and replacement, shall meet current Ministry specifications. The Ministry *Specifications for Standard Highway Sign Materials, Fabrication and Supply* are located on our Ministry web page at:

https://www2.gov.bc.ca/assets/gov/driving-and-transportation/transportation-infrastructure/engineering-standards-and-guidelines/traffic-engineering-and-safety/traffic-engineering/traffic-signs-and-pavement-markings/standard-traffic-signs/specifications/sign_fabrication_specs.pdf

The Contractor shall supply and maintain all necessary signs and traffic control devices required by the Contract, this specification, and the TMM or as ordered by the Ministry Representative. All signs and devices used must conform to the standards detailed in the TMM.

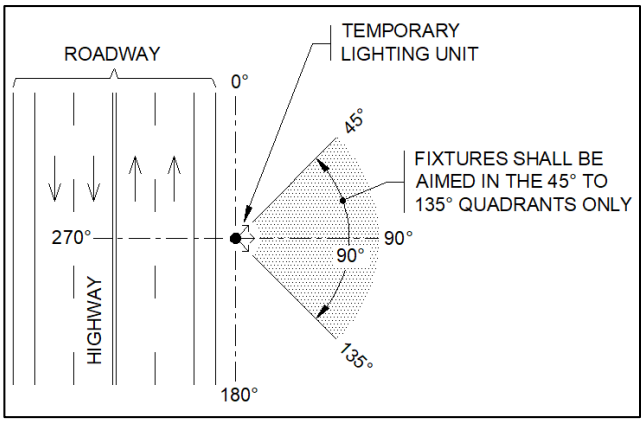
194.45 Barricades, Lights, Delineators, Temporary Pavement Markings – The Contractor shall provide, erect and maintain all types of temporary traffic control devices including temporary pavement markings, barricades, fences, delineators, amber flashing lights, in order to ensure safety to the workers and the general public. These devices shall be as required by the Contract and the TMM, or as directed by the Ministry Representative.

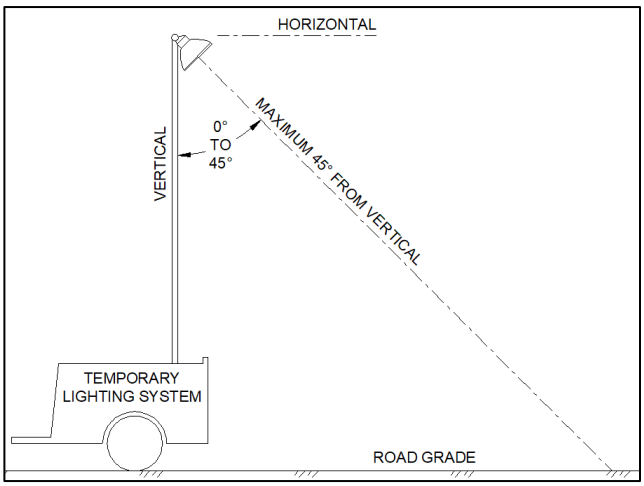
Flashing amber lights, if deemed necessary by the Traffic Control Plan, are used for extra night time effectiveness and the batteries of the flashing lights shall be checked each day.

Flashing lights, if two-faced, and if used to mark the right-hand edge of a travelled lane, must have one face covered so that the light is not visible to traffic travelling in the other direction. Flashing lights having two faces showing in opposite directions may be used only if it is necessary that the obstruction or lane edge being marked be seen from both directions of travel. They should not be used in a line of delineators through an area of incomplete construction except where traffic must be diverted and a specific hazard exists.

Temporary road construction lighting installations shall be positioned as shown in Figure 194-1.

Figure 194-1: Positioning of Temporary Lighting Adjacent to a Highway





To minimize glare and resulting impaired driver vision, lights shall be aimed away from any approaching traffic and at an angle of less than 45 degrees above vertical. Where lights cannot be located as shown then temporary glare screens shall be used to shield the traffic from the light source. These glare screens shall be metal or an opaque plastic material and be designed to shield the lighting from the eyes of the driver.

The Contractor shall be responsible for the application and removal of all temporary pavement marking and traffic control devices. When traffic lanes have to be redefined for Long Duration Work, the Contractor shall eradicate all redundant temporary or pavement markings that are not required for the intended traffic patterns.

Class I and Class II standard barricades and temporary delineator posts are illustrated in the TMM.

194.46 Dynamic Message Signs – The use of dynamic message signs (DMS) may be specified in the Contract or by the Ministry Representative. Each sign must be portable. When in operation, the DMS shall be a minimum of two (2) metres from the bottom of the sign display to the road surface and shall be level and capable of pivoting for sighting purposes. The specifications of the sign are as indicated in Table 194-A.

A full matrix sign may be used given that it has the display parameters noted above.

The Contractor is responsible for the DMS including but not limited to bringing the signs to the Project, moving and setting-up, providing storage as required including moving the signs into and out of storage, maintaining the signs in full operating condition, ensuring the correct message is being displayed, damage to or loss of the signs and returning the signs to the supplier. The Contractor shall provide the Ministry Representative with a description of the messages displayed and procedures for changing DMS messages when the Contractor is on and off site.

Table 194-A: Dynamic Message Sign Specifications

Element	Specification
Sign unit:	Yellow/Orange LED display
Sign display:	3 lines with 8 characters per line
Character size:	450 mm (18 inches)
Character matrix:	5 x 7
Remote dial-up access:	By cellular phone or equivalent

PART E – GENERAL MAINTENANCE

194.51 Maintenance – The Contractor shall be responsible for the continuing maintenance of all traffic control devices in use as follows:

(a) Signs, etc., not applying to existing conditions shall be removed or covered. If covered, such as after sundown or during periods when work is suspended, the covering shall be placed over the sign so as to completely obscure the message thereon. Where operations are carried out in stages, only those devices or signs, which apply to the current stage, are to be left in place or uncovered.

(b) Signs shall be checked daily for legibility, damage, suitability and location. Dirty, damaged, unsuitable or misplaced signs shall be cleaned, repaired or replaced. Signs and delineators with a reflective surface shall be cleaned as frequently as necessary to ensure full reflectivity. As considerable damage may be inflicted on signs and devices during a weekend, the Contractor shall conduct a check for replacement or repairs before the rush hour on Monday morning. Similarly, high traffic volume and recreational routes, and any site that suffers repeated damage, shall be checked during the weekend or at any other time that works is not in progress.

(c) During hours when headlights are required, signs that are not retroreflective shall be illuminated. After sundown all signs shall be checked for visibility and those that cannot be seen clearly shall be adjusted or replaced.

(d) Channelizing devices critical to the guidance of road users through the work zone shall be checked daily, or more frequently as specified by the Contract or Ministry Representative, for proper placement.

PART F – PAYMENT

194.91 Payment – The cost of all Traffic Management requirements, including the construction, staging, maintenance, and demolition of detours, and provision of pilot cars, all flagging and traffic control equipment, and personnel covered in this Section and the Contract Documents, or as ordered by the Ministry Representative, shall be considered to be included within the Lump Sum bid for Traffic Management, unless otherwise specified in the Special Provisions.

(a) Payment for Detour Construction – Where the Contract specifies that detour construction will be paid, such payment shall be as follows, based on the detour Design (whether prepared by the Ministry or by the Contractor), staging plan, and demolition plan, all as approved by the Ministry Representative:

(i) Contractor-Designed Detour – Any detour designed by the Contractor will be paid only where payment is explicitly specified in the Special Provisions or Schedule 7 and only as a Lump Sum, and must:

(A) be compliant with the Contract,

(B) be of reasonable scope, in the opinion of the Ministry Representative, and

(C) be approved by the Ministry Representative.

(ii) Lump Sum – If payment is specified as a Lump Sum, the Lump Sum price bid shall be full compensation for costs to construct, stage, and demolish all detours. Progress payments shall be determined by the Ministry Representative, in consultation with the Contractor.

(iii) Unit Prices – If payment is specified as being at the Unit Prices, quantities of Work required will be paid under the applicable Unit Price Items bid, measured and paid for each movement of material for construction, staging, and demolition of the detour. Where Work is required that is outside the scope of the existing Unit Prices, those costs shall be incidental to the Work and will not be paid separately.

(iv) Full Costs – In all cases, the payments made under SS 194.91(a)(ii) and SS 194.91(a)(iii) shall be full compensation for any and all Work necessary to complete the construction of the detour, stage any changes required to the detour configurations, and to demolish the detour.

(v) Detour Maintenance – Payment for maintenance of any detour, whether detour construction is paid or incidental, will be included within the Lump Sum paid for Traffic Management.

(b) Payment for Pilot Cars and Traffic Control Persons – Where pilot cars and traffic control persons are included as bid items in the Contract Documents, a record of the days and hours that each pilot car and/or traffic control person was employed shall be submitted daily by the Contractor for verification and approval of the Ministry Representative. In such event the overtime "hours worked" by traffic control persons in accordance with the current union shall be expressed in equivalent straight time "hours earned" for payment at the rate bid. Pilot cars will be paid for at the all-found rate bid in the Schedule for the actual number of hours operated as such.

SECTION 200
CLEARING AND GRUBBING

200.01 Clearing – Except for trees and shrubs to be preserved as indicated on the Drawings or designated by the Ministry Representative, "Clearing" shall mean the complete removal and disposal of all standing and fallen trees, stumps, logs, upturned roots, rotten wood and all other vegetation growth, and accumulations of rubbish of whatsoever nature, and any other objectionable material from the area shown on the Drawings, or as directed by the Ministry Representative. Protective measures for the protection and retention of vegetation are specified in SS 769.

Unless otherwise specified in the Contract, Clearing shall also include the removal of existing cribwork, guard rail, fences, siphon structures built of timber and/or metal, culvert pipes of all types and timber box culverts within the limits of the work.

Clearing shall not include the removal of concrete buildings or their foundations, concrete bridge piers or abutments and their footings, concrete siphon boxes or concrete box culverts. These items shall be classified as Type A (Solid Rock) and will be paid for as such in accordance with the measurements established by the Ministry Representative, and, where applicable, at the unit price bid in the Schedule of Approximate Quantities, and Unit Prices for Type A (Solid Rock). If the Contract does not have a Type A Item, the associated work will be paid as a Change to Work under GC 38.00.

200.01.01 Area – The area(s) to be cleared and/or grubbed shall include those areas required for the highway right-of-way, Ministry provided borrow pits, and quarry sites, Ministry or Crown gravel pits and access roads thereto, if ordered by the Ministry Representative; off-take ditches or other drainage ditches and channels for stream diversions either within or without the right-of-way; clearing of debris in streams as indicated on the Drawings, or as directed by the Ministry Representative within the Site; foundations for structures and the right-of-way as shown on the Drawings, or as directed by the Ministry Representative, for intersecting public and private road approaches or diversions of same outside the main highway right-of-way. All such areas are hereinafter referred to as the "Area."

The Contractor shall stake the limits of clearing and grubbing as defined on the Drawings. Prior to commencing the clearing and grubbing operation the Contractor shall notify the Ministry Representative of the areas to be worked. The Ministry Representative may adjust the limits of clearing and grubbing and adjust the payment accordingly.

200.01.02 Timing – Clearing and Grubbing shall only be undertaken outside the bird nesting window identified for the Site to avoid contravention of environmental legislation (including without limitation: the Migratory Birds Convention Act, the Migratory Birds Regulations 2022, Section 34 of the Wildlife Act), unless an appropriately qualified professional conducts a survey to determine there are no occupied bird nests. The survey shall be the responsibility of the Contractor and shall be performed to the satisfaction of the Ministry Representative. If required, the survey shall be considered incidental to the Work and no additional payment will be made.

Clearing must be carried out and completed in a timely fashion, to a distance and width as necessary to facilitate utility relocation and grading operations.

200.01.03 Merchantable Timber

(a) (a) License to Cut

(i) Ministry Obtained – Unless the Contract states otherwise, authorization from the Ministry of Forests (Forestry) to cut and remove trees from the Project limits will be obtained by the Ministry. Once this authorization is approved, Forestry will then issue the License to Cut to the Contractor prior to the commencement of the Clearing and Grubbing operation.

(ii) Contractor Obtained – When specified in the Contract, it shall be the Contractor's responsibility to obtain a License to Cut and Remove Timber from the Ministry of Forests prior to the removal of merchantable timber from within the Project limits. The Contractor shall submit a copy of the License to Cut and Remove Timber to the Ministry Representative prior to commencing any clearing operations.

(iii) Compliance – The Contractor shall fully comply with all Ministry of Forests and the License to Cut.

(b) (b) Merchantable Species – Merchantable timber shall be as defined by the Ministry of Forests. It shall be the Contractor's responsibility to determine which species the Ministry of Forests will class as merchantable.

(c) (c) Ownership – Merchantable timber in areas of the highway right-of-way will become the property of the Contractor and shall be removed by the Contractor in accordance with the Forest Act.

(d) (d) Processing – All merchantable timber shall be processed according to industry standards. The Contractor can be charged by the Ministry of Forests for any merchantable timber willfully damaged or destroyed during construction operations. Any fines

resulting from non-compliance of the license requirements by the Ministry of Forests shall be the responsibility of the Contractor and as such, will be deducted from the Contractor's progress payments until proof of payment has been presented to the Ministry Representative.

200.01.04 Vegetation Retention – All trees shall be felled within the Area but, in the event of any trees falling outside the Area, such trees shall be cut up and together with all debris and slash therefrom, disposed of in accordance with the Contract.

The Ministry Representative may designate certain trees or shrubbery to be left standing, in which case the Contractor shall take every precaution not to damage or injure such trees or shrubbery in felling adjacent timber, burning or any other clearing operations as specified in SS 769.

Such trees or shrubbery are to be limbed or thinned to such height and extent as may be directed by the Ministry Representative. No deduction will be made in the total quantity of clearing for any trees or shrubbery ordered to be left standing; no additional payment will be made for any limbing or thinning of such trees or shrubbery as may be directed by the Ministry Representative.

200.02 Disposal of Material

200.02.01 General – Unmerchantable timber, stumps, etc., shall not be disposed of by pushing outside the clearing and grubbing right-of-way.

200.02.02 Chipping – Organic debris from clearing and grubbing, including stumps, may be reduced by chipping, grinding or other means, mixed with the organic stripping, and disposed of in accordance with the following conditions:

(a) The organic debris has no dimension exceeding 75 mm,

(b) The organic debris is uniformly mixed with the unsuitable material at a rate not exceeding 30 parts total organic debris to 100 parts organic stripping material by volume,

(c) Material shall either be disposed of in the designated disposal area or stockpiled for future use at the direction of the Ministry Representative, and

(d) The resulting mixture is spread only in locations and to depths approved by the Ministry Representative.

200.02.03 Burning – Where permitted, any burning shall be carried out subject to the Provisions of the Open Burning Regulations and the Forest Act and regulations thereto.

Where burning is permitted, associated slash and debris shall be piled and burned at points located centrally in the Area.

The number of fires to be started at any one time shall be limited to the capacity of the Contractor's equipment and organization to provide adequate protection against the spreading of the fires to adjacent timber or property.

In the event of Forestry regulations prohibiting burning, the slash, debris, etc., may be piled along the sides of the Area and burned at a later time when permitted by the regulations. Where the regulations do not allow burning prior to the Completion Date, the material shall be disposed of off-Site prior to the Completion date.

200.02.04 Off-Site Disposal – Any debris from clearing and grubbing destined for off-Site disposal shall be disposed of at an-off-Site disposal site provided by the Contractor, in accordance with SS 145.27.02, or at other locations as permitted under the Contract.

200.03 Grubbing – Grubbing means the entire removal and disposal by burning as described in SS 200.01 "Clearing" and SS 200.02 "Disposal of Material" of all stumps, roots and embedded logs to a depth of 0.6 m below the ground line.

Grubbing shall be carried out over the entire Area as defined in SS 200.01 "Clearing." Grubbing shall be fully completed at least 300 m in advance of grading operations.

200.04 Danger Trees – This item shall consist of removing, on written order of the Ministry Representative, any leaning or dangerous trees or snags outside the clearing Area.

Such trees shall be felled and disposed of in accordance with SS 200.01 "Clearing" and SS 200.02 "Disposal of Material."

The Contractor will not be required to grub the stumps of any danger trees.

200.05 Mailboxes – Where it becomes necessary to remove or otherwise disturb existing mail boxes within the limits of any project, the Contractor will be required to install the boxes temporarily in such a position that their usefulness will not be impaired. After construction work has been completed, the boxes shall be re-installed at positions designated by the Ministry Representative. No new posts or materials will be required, except that any damage caused by the Contractor to either boxes or supports, due to the Contractor's negligence, shall be repaired by the Contractor at the Contractor's expense.

All costs in connection with the removal and replacement of the mailboxes, as specified above, shall be considered as incidental to the construction of any improvement and shall be included in the unit prices for the various pay items of work involved in the Contract. No additional compensation will be made.

200.06 Removal of Existing Signs and Symbols – The Contractor shall remove all existing signs and their associated structures as shown on the Drawings and, where applicable, in accordance with SS 635.

The Contractor must obtain approval from the Ministry Representative prior to removing any signing equipment

and, if so directed by the Ministry Representative, existing signs shall not be removed until new signs or temporary signs have been installed.

The Contractor shall remove and stockpile all existing sign faces, boards, redundant posts and bases at the Ministry Maintenance Contractor's closest yard, or as otherwise directed by the Contract or the Ministry Representative.

Voids remaining after the removal of sign bases shall be backfilled and compacted with suitable native materials.

Existing Ministry signs, which are not marked for removal or relocation but are in conflict with the progress of the Project, shall be removed and temporarily relocated as approved by the Ministry Representative. No separate payment will be made for these temporary relocations which shall be considered incidental to the Works and no additional payment will be made.

MEASUREMENT AND PAYMENT

200.91 Clearing – CLEARING will be paid for at the unit price bid per hectare of clearing and will, unless otherwise specified, comprise the full length and breadth of the right-of-way as shown on the Drawings or as directed by the Ministry Representative. The price shall be accepted as full compensation for everything furnished and done in connection therewith, regardless of the nature or condition of the Area.

200.92 Merchantable Timber – All costs associated with sorting, loading hauling and cold-decking, and disposal of merchantable timber, including but not limited to, the provision of the maintenance of the cold-deck sites and the construction and maintenance of haul routes and other roads shall be considered incidental to the Work and no additional payment will be made. No separate payment will be made in respect to the time taken to remove the timber from the cold-deck sites, and all costs resulting from the presence of timber in the cold-decking sites, until such time as it may be removed, including but not limited to, the cost of property, of scheduling the Work, of performing additional Work, and of delay in the Work.

200.93 Grubbing – GRUBBING will be paid for at the unit price bid per hectare of grubbing and will, unless otherwise specified, comprise the full length and breadth of the right-of-way as shown on the Drawings or as directed by the Ministry Representative. The price shall be accepted as full compensation for everything furnished and done in connection therewith, regardless of the nature or condition of the Area.

200.94 Danger Trees – DANGER TREES will be measured in three sizes according to the diameter of the tree trunk measured at a point 1.25 m above the ground line at the base of the tree, with compensation for removal paid in accordance with Table 200-A.

Table 200-A: Danger Tree Removal Compensation

Tree Diameter (m) @ 1.25 m above ground	**Compensation**
< 0.75	$140.00
0.75 to 1.5	$330.00
>1.5	$550.00

200.95 Remove Existing Signs – Payment for REMOVE EXISTING SIGNS including, but not limited to, excavation, removal, hauling, disposal of sign faces, removal, and stockpiling or disposal of existing posts and bases, and backfilling the excavation will be made at the Unit Price bid per sign removal and shall be considered full compensation for everything furnished and done in connection therewith. Signs without posts (pole-mounted, etc.) shall be considered the same as signs with posts for payment purposes.

No separate payment will be made for the removal and disposal of delineator posts which shall be considered incidental to the Works and no additional payment will be made.

SECTION 201

ROADWAY AND DRAINAGE EXCAVATION

DESCRIPTION

201.01 General – Roadway and Drainage Excavation shall include:

(a) all excavation and the construction of all embankments required for the formation of the roadbed, pullouts, parking areas and look-outs;

(b) excavation for any drainage ditch, off-take ditch or channel for stream diversion within or outside the right-of-way;

(c) removal of surcharge material, topsoil and organic waste material from the roadway, and of topsoil and unsuitable overburden from any Ministry provided borrow pit or any available Ministry or Crown gravel pit;

(d) excavation of materials below grade; excavation of borrow pits; the grading necessary to construct any frontage road, gravel pit access road, borrow pit access road or any other access road, public or private, either within or outside the right-of-way, only to Ministry provided sources as may be ordered by the Ministry Representative;

(e) the grading of connections to intersecting roads, public or private, either within or outside the right-of-way;

(f) the hauling and disposal of all excavated material, and the trimming and shaping of all excavations and embankments.

All Works shall be confined to the right-of-way except where agreements for access rights are in place for other lands, public or private.

201.02 Extent of Work – The dimensions of the excavations and embankments shall be as shown on the Contract Drawings but the Ministry Representative may increase or decrease roadbed dimensions and slope angles.

Construction Grade is defined as the grade upon which the Asphaltic Pavement is laid.

Subgrade is defined as the grade upon which the first layer of select granular subbase or base material is laid.

MATERIALS

201.11 Description of Material Types – Excavation shall be classified by the Ministry Representative, under the following material types:

(a) Type A – Solid Rock

(b) Type D – Common, with potential sub-classes

(i) Topsoil

(ii) Stripping

(iii) Suitable

(iv) Unsuitable

The material encountered in any project excavation shall be classified as one of the two types listed above.

In no case shall a material be classified using percentages of two or more types.

201.11.01 Type A - Solid Rock – Type A shall, without limitation, include all forms of "solid rock in place" including solid formations, masses, ledges, seams or layers of dense sedimentary, igneous or metamorphic material of sufficient hardness generally requiring drilling and blasting methods, very heavy ripping, or equivalent methods, before excavation and removal.

Type A shall also include detached masses of rock or boulders individually containing a volume of 2.0 m^3 or more.

The breaking and removal of frozen materials shall not be considered Type A excavation.

201.11.02 Type D – Common – Common material is all other excavation materials of a nature not included in the foregoing description of Type A, regardless of the nature or condition of the material, or the method used to excavate or remove. Type D may be broken down into further sub-classes for payment, generally depending on handling requirements and intended use:

(a) **Topsoil** – typically the upper organic soil layer consisting of intermixed humus, minerals and composted materials, well suited as a growth medium for plants and unsuitable for embankment construction. Used only where planting beds require higher quality growth medium.

(b) **Stripping** – Stripping is a sub-class of Type D that is material typically near or at the surface and contains organics, root mass, or vegetation., which is typically unsuitable for embankment construction but may be suitable for slope dressing. Mulched materials, depending on the size of the chips and proportion of soil to chips, may be considered as Stripping. Where a Contract has no specific use for Topsoil, Topsoil may be included within Stripping

(c) **Suitable** – material suitable for embankment construction.

(d) **Unsuitable** – material which is unsuitable for embankment construction but is not Stripping or Topsoil. May include mulching, where chip sizes are large or soil to chip proportion is low.

201.12 Material Classification Changes – It is possible that the material in a cut may change at some level in the excavation and may change more than once in a single cut.

Where such a change occurs, the Contractor shall immediately, and in any event within twenty-four (24) hours of attempting to excavate the changed material, notify the Ministry Representative and clean off any material falling within the currently approved material classification of the excavation, in order to expose the horizon where the material change is thought to occur.

The Ministry Representative will review the exposed material, determine whether a classification change is warranted, and notify the Contractor if a classification change is warranted or not.

If a classification change is approved, sufficient field measurements will be taken to establish boundaries and the material horizon for volume calculations.

If the Contractor disagrees with the Ministry's assessment of the material classification, the Contractor may appeal only in accordance with SS 201.13.

201.13 Material Classification Appeal

201.13.01 Type A and Type D Material Classification – The Contractor may appeal the material classification between Type A and Type D as determined by the Ministry Representative. There shall be no appeal for any distinction between the Type D sub-classes.

201.13.02 Notice and Time Limit of Appeal – The Contractor shall serve written Notice of appeal to the Ministry Representative within:

(a) twenty-four (24) hours of the Contractor attempting to excavate any potentially changed classification material; or

(b) three (3) days of the material classification designation by the Ministry Representative; or

(c) in no event less than two (2) days prior to the completion of the excavation.

201.13.03 Ripping Test – Where the Contractor appeals the material classification, a ripping test will be conducted under the direction and control of the Ministry Representative, including the determination of the location, time, and suitability of weather, ground, equipment and other conditions for the ripping test to occur.

The ripping test will be conducted using a Class 7 or more powerful bulldozer (as identified in the "Equipment Rental Rate Guide" published by the BC Road Builders and Heavy Equipment Association), that will develop sufficient traction and effectively deliver a minimum force of 700kN per lineal metre of ripper shank embedded in the material by a single shank static ripper as determined by the manufacturer's Drawbar Pull versus Ground Speed Charts.

If, under the above specified ripping force, the material can be broken or loosened into pieces with an individual volume of 2.0 m^3 or smaller, the material will be classified as Type D. If the material cannot be broken or loosened within the time limit as directed by the Ministry Representative, the material will be classified Type A.

The test will be conducted at the Contractor's expense unless the test determines, in the sole determination of the Ministry Representative, that a material classification change is warranted. In that case, the cost of the equipment (including mobilization and demobilization if the equipment is brought in solely for the purpose of the ripping test) and the operator will be paid as Extra Work.

201.14 Unsuitable Material – Only material acceptable to the Ministry Representative shall be used in the construction of embankments.

Where Topsoil is a payment Item in the Contract, Topsoil within the excavation and embankment area, and meeting the requirements of SS 751.13 and SS 751.16, shall be excavated and stockpiled at a location acceptable to the Ministry Representative. Where the Contract has no Item for Topsoil or the available volume of Topsoil exceeds that required for the Contract, only the required volume shall be stockpiled and the balance dealt with as Stripping.

Stripping and surplus Topsoil shall be removed over the entire excavation and embankment area. Topsoil shall be stockpiled separately from Stripping.

All material above or below subgrade in an excavation or in the foundation of an embankment, which in the opinion of the Ministry Representative is unsuitable for embankment construction, shall be removed and paid as classified by the Ministry Representative. Soft clays and peats are typical examples of unsuitable materials.

Material deemed unsuitable for the construction may be wasted between the toe of the embankment and the right-of-way boundary or in other areas, all as designated by the Ministry Representative. If no waste areas are designated in the Contract then the unsuitable material shall be removed and disposed offsite at the Contractor's expense. Any such material subsequently placed in an embankment, without the approval of the Ministry Representative, shall be removed and disposed of, as directed by the Ministry Representative, and no payment will be made for the removal or haul of such unsuitable material from the embankment.

Where removal of unsuitable material requires backfilling this work shall be carried out with approved fill materials, as directed by the Ministry Representative. Payment for such backfilling and haul will be made at the Unit Price bid for "Roadway and Drainage Excavation".

Topsoil shall be placed in designated planting areas, in accordance with the Contract.

Upon completion of the road grade, all Stripping, and Surplus Topsoil and material suitable for growth of ground cover shall be spread as slope dressing to a nominal depth of 150 mm on the cleared right-of-way and up the fill slopes to generally blend in with the cross section. Care shall be taken that proper drainage is maintained. This material shall be placed, compacted and trimmed to a neat appearance in a manner to facilitate growth of revegetation. No additional payment will be made for placing and trimming material used for slope dressing.

201.15 Surplus Material – All suitable excavation material shall, if required, be taken or hauled into the nearest embankment and to any embankment on the project to be constructed out to the extreme distance required by the Ministry Representative.

The surplus beyond what is necessary to form the nearest embankment shall be:

- disposed of by widening embankments uniformly within the limits of right of way;
- hauled to form or widen any embankment on the road to be constructed;
- stockpiled on-site or in Ministry pits; or
- lastly removed from Site for disposal by the Contractor,

all subject to the approval by the Ministry Representative, where hauling and placing shall be performed in the most efficient manner.

No material shall be wasted without the approval of the Ministry Representative. All surplus material shall be used as much as possible.

201.16 Contaminated Soils – If any contaminated soils, suspected or actual, are encountered during construction, the material is to be left in-situ or be segregated and stockpiled, and handled in accordance with SS 165 Appendix 165-A – Managing Suspect Contaminated Materials.

In general, the Appendix requires segregating, testing by a Qualified Environmental Professional retained by the Contractor, handling, and disposal of the excavated material at an approved disposal site or at a licensed disposal facility.

Prior to any disposal, the Contractor shall identify the disposal site(s) to the Ministry Representative, in accordance with SS 145.27.02.

Contaminated material hauled to off-Site disposal will be paid at applicable Unit Price for Contaminated Soil Disposal or, where there is no such bid Item, as Extra Work.

CONSTRUCTION

201.31 Rock Cuts – See SS 204.

201.32 Overhanging Rock and Boulders – The Contractor shall remove overhanging rock or boulders within the excavation limits, as ordered by the Ministry Representative, and shall backfill and compact all resulting cavities to allow the safe passage of construction and support vehicles generally in accordance with the requirements of SS 135. Such work will be paid at the Unit Price bid for the class of excavation involved.

201.33 Overbreak in Solid Rock - Overbreak in solid rock excavation encountered in the contract is that portion of any such material which is excavated, displaced or loosened outside and beyond the slopes or grades as staked or re-established (with the exception of slides as described in SS 201.42).

Any overbreak that occurs due to the inherent character of any formation encountered, as determined by the Ministry Representative in consultation with the Qualified Professional will be classified as allowable overbreak as described in SS 201.33.01.

All other overbreak, as so defined, shall be removed by the Contractor at the Contractor's expense, and shall be disposed of by the Contractor, in the same manner as provided for "Surplus Material", but at the Contractor's expense.

The Qualified Professional is defined as the Ministry's Geotechnical Engineer, Rockwork Engineer, or consultant engineer working for the Ministry whose discipline is in geotechnical engineering.

201.33.01 201.33.01 Allowable Overbreak – If, and only where directed by the Ministry Representative, use is made of the overbreak to replace material which would otherwise have to be excavated outside the design cut line on-Site or borrowed for the construction of the roadway, then such overbreak will be classed as allowable overbreak. Any overbreak that is not incorporated within the approved design lines and grades of the embankments shall not be considered "allowable overbreak".

201.33.02 Contractor's Negligence – Overbreak resulting from fault or negligence of the Contractor shall be removed and disposed of, as directed by the Ministry Representative, at no cost to the Ministry, subject to SS 201.33.01.

Definition of negligence will be deemed to be lack of the following: the sequence of operations, lack of a blast design, lack of preblast survey, lack of as-built blast design as described in SS 204.04.06, SS 204.04.07, and SS 204.04.08 and SS 204.04.09. Also, definition of negligence will be deemed to mean lack of cut-off drill holes on the backslope, excessive spacing of drill holes and overloading of the same.

201.34 Sealing of Grade – To prevent standing water in the cuts and fills, the Contractor shall at the end of each working

day, grade, roll and seal the materials with a smooth wheel roller weighing not less than 5,400 kg, such that the water will not pond on the grade.

201.35 Enlargement of Excavation – In cases where the quantity of material taken from a regular excavation will not be sufficient to form the required embankment, the deficiency shall be supplied by taking material from the excavation within or outside the right-of-way, at such places as the Ministry Representative may direct, or from enlargement of the regular excavations made uniformly on one or both sides.

The sides of the excavation in all cases shall be dressed to such slopes as the Ministry Representative may require. When the sides of the excavation had previously been final-dressed payment for sloping or scaling the second time will be as Extra Work.

201.36 Rock Embankments – This section applies to embankments constructed from material containing more than 15% by volume of rock larger than 150 mm.

Embankments shall be constructed in layers equal in thickness to the largest size of the material but not exceeding 0.7 m. Greater lift thicknesses will be permitted by the Ministry Representative under special conditions provided the Contractor can spread the larger material satisfactorily and attain a degree of compaction no less than attained for layers less than 0.7 m in thickness.

Regardless of layer thickness and material particle size all material shall be well compacted to achieve the design density and stability of the embankment and to the satisfaction of the Ministry Representative.

The material shall be deposited and spread so that the larger rocks are well distributed and the intervening spaces are filled with smaller sizes as may be available to form a stable embankment. The finer portion of excavated rock shall be retained for the top transition layer. Each layer shall be compacted by routing the loading construction equipment over the entire width supplemented with additional compaction equipment, as necessary to ensure compaction is uniform, a stable embankment is achieved and to the satisfaction of the Ministry Representative.

When a rock embankment is overlain by an earth embankment or by subbase or base course materials, the top 0.5 m of the rock fill shall be sealed with smaller rock particles and suitable soil materials, if necessary, to prevent the uppermost layers penetrating into voids in this rock embankment.

Where permitted by the Ministry Representative, and where the width is too narrow to accommodate equipment, material for the toe of the side hill rock fills may be placed by end dumping until sufficient width of the embankment has been formed to carry equipment, after which the remainder shall be placed in layers and compacted as specified. Any loose material left on the slopes by end dumping shall be incorporated in the fill and compacted according to these standards.

When embankments are made on hillsides or where a new fill is to be applied upon an existing embankment, the slopes of the embankment or original ground (except rock) shall be terraced in a continuous series of steps a minimum of 1.5 m wide as the embankment rises.

The material from step excavation shall be spread and compacted into the adjoining embankment. No additional payment will be made for excavation or for placing step material in the adjoining fill.

201.37 Earth Embankments – This section applies to embankments constructed from material containing less than 15% by volume of rock larger than 150 mm. Embankments shall be constructed in such a manner that they shall be completely stable with reference to designated traffic loading.

The natural surface shall be excavated to remove organic soils or other unsuitable material, as directed by the Ministry Representative, and the removal of such material will be paid at the Unit Prices bid for the class of excavation involved.

When the foundation of the embankment consists of sensitive soils the Contractor shall, with the Ministry Representative's approval, place the initial lift of embankment to a minimum depth required as stated in the design to carry the Contractor's hauling equipment. Static compaction shall be used in areas of sensitive soil.

The earth embankment shall then be constructed in successive horizontal layers and thicknesses and compacted to the required percentage of Standard Proctor Maximum Dry Density obtained by ASTM D698, all as specified in Table 201-A.

Table 201-A: Lift Thickness and Required Density by Depth Below Subgrade

Depth Below Subgrade (mm)	Maximum Uncompacted Lift Thickness (mm)	Minimum Density (% SPD)
0 to 300	100	100%
300 to 500	100	95%
Below 500	200	95%

The above maximum lift thickness requirements may be waived by the Ministry Representative, where the Contractor can demonstrate that:

(a) the proposed compaction equipment can uniformly compact a thicker layer;

(b) the equipment and compaction procedures attain the required degree of compaction at all levels within the lift;

(c) they have compaction testing equipment and services for Quality Control and Quality Assurance that will reliably test to the full depth of the proposed lift thickness. The supply of all testing equipment for Quality Control and Quality Assurance shall be the Contractors responsibility.

Where the embankment material is highly variable or of such character that conventional laboratory density determination or conventional compaction testing are likely to be inappropriate, the Contractor may prepare a proposed compaction plan based on the "Target Density" provisions of SS 202.22, roll patterns or other methodology, for review and sign-off by the Quality Manager, and submission to the Ministry Representative. The Ministry Representative, in consultation with the Geotechnical Engineer of Record, may approve or reject the proposal.

No organic soils shall be placed in the embankment. Soils with high moisture content that cannot be compacted to the required density shall not be employed without prior aeration and drying.

When embankments are made on hillsides or where a new fill is to be applied upon an existing embankment, the slopes of the embankment or original ground (except rock) shall be terraced in a continuous series of steps a minimum of 1.5 m wide as the embankment rises.

If suitable, the material from step excavation shall be spread and compacted into the adjoining embankment. No additional payment will be made for excavation or for placing step material in the adjoining fill.

The Contractor shall be responsible for selecting equipment and methods of attaining the specified degree of compaction. In general, the roller mass shall be sufficient to compress the soil vertically after each pass but not sufficient to unduly rut or shear the soil.

Water shall be added and incorporated into the soil using suitable equipment such as rotary mixers, cultivators, etc., to increase the natural moisture content to the optimum moisture percentage as determined by the current ASTM D698.

In the event that the natural moisture content is greater than the optimum, the soil shall be aerated and dried employing suitable mixing equipment.

201.38 Frozen Material – The use of frozen material in embankments will not be permitted; the only exception is, with Ministry Representative approval, broken rock containing less than 15% passing a 4.75 mm sieve. Frozen excavated material which will be suitable when dry shall be stored and allowed to thaw and dry, and then placed in the embankment, as directed by the Ministry Representative. No compensation will be allowed for the storing and re-handling of this material.

No material is to be placed on a frozen surface unless approved by the Ministry Representative.

201.39 Snow Removal – Snow overlying the surface of a cut, or the site or surface of an embankment shall be removed and deposited beyond the slope stakes at the Contractor's own expense.

201.40 Bridge End Fill – Material for bridge end fill shall be in accordance with SS 202.04 and SS 202.05. Construction of bridge end fill shall be in accordance with SS 202.23.

201.41 Transition Sections – Subgrade, other than solid rock, shall be subcut 1 m deep at the line of transition from cut to embankment. The subcut shall taper to zero depth 8 m within the cut. The embankment construction shall be carried back over the subcut only after the embankment has reached the level of the bottom of the subcut, as measured by the Ministry Representative. Payment for the subcut will be made at the Unit Price bid for the class of excavation involved

201.42 Slopes and Slides – The slopes of all excavations and embankments shall be trimmed neatly and evenly to the line and slope indicated on the Drawings, including special slope treatment in accordance with SS Drawing SP201-01, or as directed by the Ministry Representative.

No undercutting of slopes in excavation by power shovels or other excavation equipment will be permitted.

In case slopes, finished to the lines as shown on the Drawings, slide into the roadway or out of embankments before final acceptance of the work, such slide material shall be removed by the Contractor from the roadway or replaced by the Contractor in the embankment, as the case may be, at the Unit Price bid for the class of excavation involved. The classification of material in slips and slides shall be in accordance with its condition at the time of removal regardless of its prior condition. The slopes shall be refinished by the Contractor, as directed by the Ministry Representative. Such refinishing will be paid for by Order for Extra Work. Materials to replace embankment slides shall be obtained from sources designated by the Ministry Representative.

Slopes undercut at the base or destroyed in any manner by act of the Contractor shall be resloped by the Contractor at the Contractor's expense to the slope, as directed by the Ministry Representative. All materials resulting from such resloping shall be removed and deposited, as directed by the Ministry Representative. No payment will be made to the Contractor for the removal of such material.

201.43 Ditches – Ditches which may be considered necessary for the proper drainage of the work shall be constructed at such points and to such cross section, alignment and grade as the Ministry Representative may

direct. This shall include inlets and outlets to culverts and ditching of all kinds. Ditching quantities will be considered as ordinary excavation quantities and will be measured, classified and paid for as such.

201.44 Borrow – Borrow refers only to materials sourced outside of the Site; Material sourced on-Site, within or outside of the design lines, is Type A or Type D excavation.

Borrow shall consist of rut resistant material, with less than 20% passing the 0.075 mm sieve and free of organics, high plasticity clays and other unsuitable materials, obtained from an approved source of supply (e.g. pit or quarry) outside the highway right-of-way, developed and used in accordance with SS 202.

The Special Provisions may identify potential sources, additional or different material qualities, or require representative sampling and testing of the material.

201.45 Haul – "Haul" includes all work necessary to move excavation materials from their in-situ source to any destination, on or off-Site.

201.46 Watering – Water for compacting embankments, constructing subgrades and surfacing, and for laying dust caused by grading operations and traffic, shall be applied as required. Water shall be applied by sprinkling with tank trucks equipped with spray bars and suitable control apparatus.

The Contractor shall make all necessary arrangements for obtaining and applying water, including acquiring all required permits.

Watering shall be incidental to the Work.

201.47 Finishing of Roadway – Before acceptance and final payment is made, the entire roadway, including the roadbed, shoulders and ditches shall be neatly finished and trimmed to the lines, grades and cross sections shown on the Drawings, or as directed by the Ministry Representative, to reproduce smooth surfaces, slopes and a uniform cross section. Subgrade shall be finished within a tolerance of ± 15 mm except for rock cut fills where the tolerance shall be ± 50 mm.

All drainage ditches, waterways and culverts shall be opened up and cleaned out to restore them to their full effectiveness.

All loose rock and boulders within the right-of-way resulting from grading and grubbing operations shall be gathered up and buried, or otherwise disposed of as the Ministry Representative may direct.

The Contractor shall grade all portions of the right-of-way outside the Excavation and Embankment areas to conform to the general ground line. This work will be considered as subsidiary work pertaining to the item of "Roadway and Drainage Excavation," and no extra payment will be made.

201.48 Surcharging – Pre-consolidation by surcharging shall be carried out where indicated on Drawings and/or Special Provisions, or as directed by the Ministry Representative. Embankments on these areas shall be placed directly on the natural ground without removal of the organic materials, unless otherwise directed by the Ministry Representative, and shall be constructed to a surcharged height above construction grade as shown or specified.

The Contractor shall place the initial lift of embankment to a minimum depth required to carry the Contractor's hauling equipment with the approval of the Ministry Representative. Static compaction shall be used in areas of sensitive soils. The remainder of the embankments shall be constructed in accordance with procedures set forth in this Section; except that compaction will not be required on the surcharge material above construction grade.

To avoid shear failures, the rate of construction shall be rigidly controlled by instrumentation installed by the Ministry.

The surcharge shall remain on the embankments for a period of time as outlined in the Contract Special Provisions, or as may be indicated by the instrumentation, or as determined and adjusted by the Ministry Representative.

All instrumentation shall be kept in working, continuous, and operable order according to manufacturer's requirements.

Should any of the instrumentation become damaged or rendered unusable by the construction operations, the same will be replaced by the Ministry at the Contractor's expense.

201.49 Backslope Stabilization – Drain holes, rock bolts, mesh and/or shotcrete may be required where rock is being excavated.

The backslope stabilization requirements will be assessed by the Ministry Representative during construction and the amount and location of the drain holes, rock bolts, mesh and shotcrete may be changed or deleted according to the condition of the rock encountered in the field.

MEASUREMENT

201.81 Borrow – Borrow quantities will be measured in-bank at the off-Site source and computed in accordance with SS 145.21.01. The volume of boulders and unsuitable material removed from borrow pits that is not used in embankments shall be deducted.

201.82 Excavation – All "Roadway and Drainage Excavation" will be measured in-bank at the source, after Clearing and Grubbing operations have been completed, as specified in SS 200.01 through SS 200.03, and volumes computed in accordance with SS 145.21.01.

Pay quantities will be computed, in accordance with SS 145.21.01, in CUBIC METRE to the neat lines staked.

Adjustments for curvature will be made in any cut where deemed equitable by the Ministry Representative.

201.83 Haul and Overhaul – Haul and Overhaul are incidental to the Work.

PAYMENT

201.91 Borrow – Payment for BORROW materials, imported from outside the Site, shall be at the Unit Price per cubic metre bid for the various types of material borrowed, classified in accordance with SS 201.11. All specifications in this Section relating to excavation shall apply to borrow. Payment shall be full compensation for everything furnished and done, including without limitation costs for acquisition, development, and payment of royalties for private pits or quarries, purchase price of borrow material, access road construction and maintenance, excavation, screening, crushing, stockpiling, loading, hauling, spreading, compaction and moisture adjustment (watering or drying) in place as specified.

201.92 Excavation – Payment for EXCAVATION, within the Site, shall be at the Unit Price bid per cubic metre for the various types of materials excavated, classified in accordance with SS 201.11. The Unit Price for such excavation shall include excavation, slope trimming, hauling, and placing and compacting into embankment, stockpile or disposal site and shall be accepted as full compensation for everything furnished and done in connection therewith.

Material will be paid only once as a class or sub-class of material. For clarity, if a Contract includes Items for both Type D and its subclass Topsoil, Topsoil will only be paid under the Topsoil Item and not also as Type D.

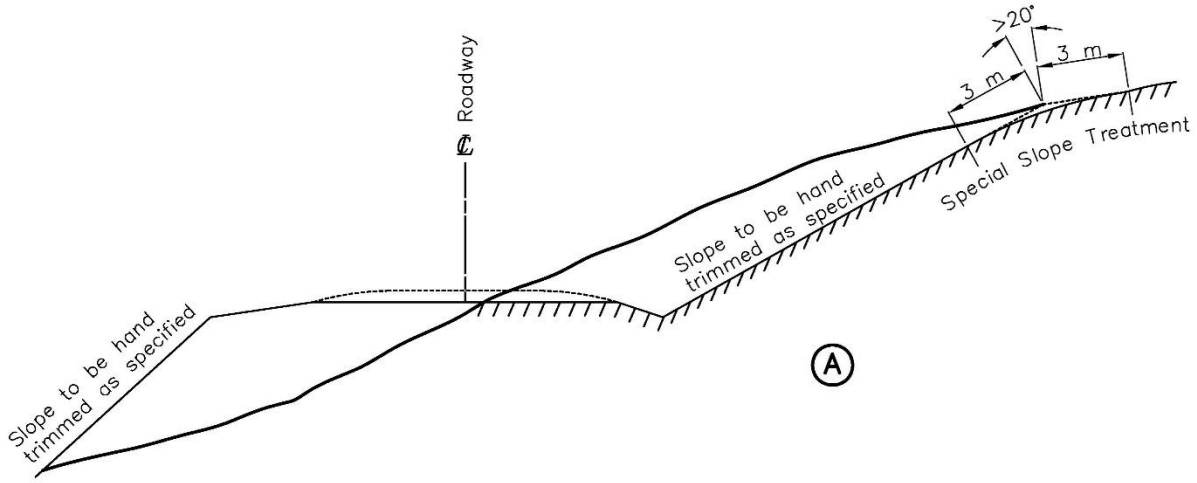
201.93 Allowable Overbreak – Allowable overbreak will be paid at 75% of the Unit Price bid for TYPE A MATERIAL. The quantity at the Unit Price for overbreak in any one cut shall not exceed 10% of the original theoretical cut, as shown on the Drawings or as directed by the Ministry Representative. Material in excess of the above 10%, if used in lieu of borrow and only on the advice of the Ministry Representative, will be paid at the Unit Price bid for TYPE D MATERIAL.

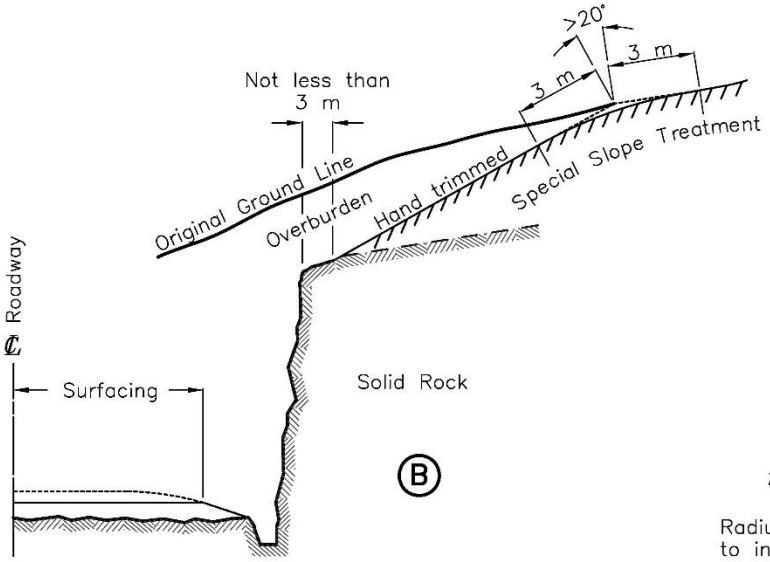
201.94 Embankment – Except where otherwise explicitly provided in the Contract, the work of embankment construction will not be paid for directly as a pay item, but shall be considered as incidental to the classes of EXCAVATION or BORROW.

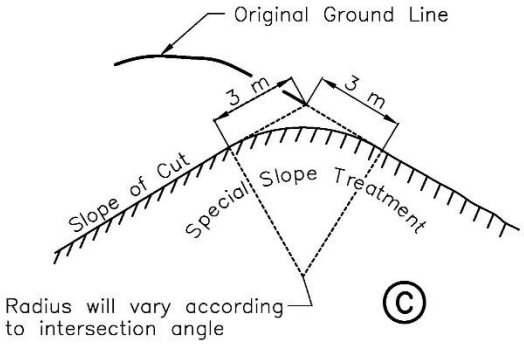
201.95 Surcharge Removal – Payment for surcharge removal shall be at the Unit Prices bid for EXCAVATION.

SPECIAL SLOPE TREATMENT

SP201–01







NOTES:

1. SPECIAL SLOPE TREATMENT REFERS TO THE ROUNDING OFF AT THE INTERSECTION OF CUT SLOPES WITH THE EXISTING GROUND LINE. IT SHALL BE APPLICABLE WHERE THE MATERIAL IS OTHER THAN SOLID ROCK, AND WHEN THE DIFFERENCE IN THE INTERSECTION ANGLE BETWEEN THE BACKSLOPE AND THE ORIGINAL GROUND EXCEEDS 20 DEGREES.
2. MATERIAL REMOVED IN SPECIAL SLOPE TREATMENT SHALL BE UTILIZED OR DISPOSED OF IN ACCORDANCE WITH **SS 201.14** AND **SS 201.15**

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

SECTION 202

GRANULAR SURFACING, BASE, AND SUBBASES

SCOPE OF SPECIFICATION

202.01 Preliminary and General – This specification describes the classification, the materials and the professional standards required for the production of granular aggregates and the construction of granular surfacing, granular base and select granular subbase.

MATERIALS

202.02 Aggregate Classification

202.02.01 Crushed Surfacing Aggregates:

(a) High Fines Granular Surfacing Aggregate (HFSA): HFSA is used for surfacing of gravel roads. Cohesion for this aggregate is achieved by plastic fines.

202.02.02 Crushed Base Course Aggregates – These are high strength, high load bearing, high quality, crushed granular base materials. As these materially are not well suited for long-term traffic, they are usually overlaid by surface course materials (HFSA or asphalt/concrete pavement).

(a) Well-Graded Base (WGB): WGB is the standard base course material used for pavement structures. It is well-graded with moderate permeability.

(b) Intermediate-Graded Base (IGB): IGB is intermediate in gradation and permeability between well-graded aggregate and open-graded aggregate.

(c) Open-Graded Base (OGB): OGB contains a small percentage of fine-grained material and is highly permeable, gap-graded aggregate.

202.02.03 Subbase Aggregates – A granular material placed between the sub-grade and the base course aggregates, providing strength and drainage to the pavement structure.

(a) Select Granular Subbase (SGSB): SGSB is the standard aggregate, crushed or screened, placed directly above subgrade, with moderate permeability and a wide gradation.

(b) Intermediate Graded Subbase (IGSB): IGSB is a higher permeability, crushed subbase aggregate.

(c) Open Graded Subbase (OGSB): OGSB is a highly permeable, crushed subbase aggregate.

202.02.04 Bridge End Fill (BEF) – BEF is quality granular fill primarily used behind and below a bridge abutment to provide good drainage, a smooth transition from the bridge approaches to the bridge structure, and a suitable material through which to drive piles.

202.03 General Description of Work – The aggregates for granular surfacing, base and subbase may be supplied either by the Contractor or produced from a Ministry provided source. Production of granular surfacing and granular base requires crushing and placing in stockpile. Production of select granular subbase and bridge end fill may be by crushing, screening or direct excavation.

Granular aggregate shall be loaded from the stockpile, crushing and screening plant, or pit face, as may be applicable; hauled out to the site, placed accurately without segregation and thoroughly compacted to established line and grade.

The Contractor shall maintain the prepared surface of the base course aggregate until it has been paved or until completion of the contract, whichever is applicable.

202.04 Aggregate Quality

202.04.01 General – Granular aggregate shall be composed of inert, clean, tough, durable particles of crushed rock, gravel, sand and fines capable of withstanding the deleterious effects of exposure to water, freeze/thaw, handling, spreading and compacting and design traffic loading. The aggregate particles shall be uniform in quality and free from clay lumps, and wood.

Aggregates for surfacing, base, subbase and bridge end fill shall be tested in accordance with the tests listed in Table 202-A.

202.04.02 Primary Quality Tests – All aggregates for surfacing, base, subbase, and bridge end fill shall meet the requirements of Table 202-B.

Where the Ministry has pre-existing test results, the Ministry will make this information available to the Contractor and the Contractor will be able to rely on the factual data and not have to perform the corresponding primary quality tests on materials extracted from the investigated portions of the pit or quarry.

In circumstances where historic performance of material from a source has proven satisfactory, the Ministry may, by Special Provision, waive any or all of the testing required in SS 202.04.

202.04.03 Secondary Quality Tests – All aggregates for surfacing, base, subbase, and bridge end fill failing any primary quality test indicated in SS 202.04.02 shall further be tested as follows to determine acceptability or, at the Contractor's option and at the Contractor's expense, a new source shall be found.

(a) If the Micro Deval test results show that the material fails, then the soundness shall be reassessed by use of Magnesium Sulphate (ASTM C88). The material will

be considered acceptable if the loss after 5 cycles is less than:

(i) 20% for coarse aggregate (the material retained on or above the 4.75 mm sieve), and

(ii) 25% for fine aggregate (the material passing the 4.75 mm sieve).

(b) If the aggregate fails any of the Sand Equivalent, Micro-Deval or Magnesium Sulphate tests, then the Contractor shall perform the Ministry's petrographic test (APPENDIX 202-A) to determine the cause of failure. The Ministry Representative may review the test results and determine whether the material is acceptable or not.

202.05 Aggregate Gradation – Gradation shall be determined in accordance with ASTM C117 (Wash Test).

202.05.01 Gradation Curve Slope – Aggregate shall have a gradation that defines a curve (% passing versus log sieve size) with a slope between adjacent sieves, equal or intermediate to the corresponding slopes of the boundary curves defined by the specification. Gradations shall fall within the limits, for the specified classification, shown in Table 202-C.

202.05.02 Rut Resistance – Any aggregate supplied must, in addition to meeting the gradation requirements specified above, not rut when proof rolled with a truck having a 9 tonne single axle dual tire or 17 tonne tandem axle group with dual tires with a tire pressure of 600 kPa. Any aggregate which does rut shall be removed and replaced, or blended with suitable aggregates, to meet both the gradation requirement and the rut resistance requirement.

USE OF MINISTRY-PROVIDED PITS OR QUARRIES

202.06 General – Ministry-provided pits and quarries are those identified in the Special Provisions which the Ministry is offering to the Contractor as potential aggregate sources.

They may include pits and quarries:

(a) owned by the Ministry

(b) leased by the Ministry, or

(c) established as Map Reserve on Crown Land.

Use of such pits and quarries will be at the Contractor's option unless the Special Provisions clearly state that the source must be used for specific purposes.

When a Ministry pit or quarry is provided for the Contractor's optional use under the Contract, the Contractor, by entering the pit or quarry, will be deemed to have accepted all obligations, risks and costs for the production of aggregate meeting the requirements of SS 202.03, SS 202.04, and SS 202.05. The Ministry gives no warranty that its pit or quarry will meet the quality and quantity requirements.

Where a pit or quarry is designated as a mandatory source, the Ministry waives the aggregate quality requirements of SS 202.04.02 except for the Fractured Faces requirement for aggregates required by the Ministry to be produced from that source.

Table 202-A: Aggregate Quality Tests

ASTM	**Title of Test** Standard Test Method for…
C88	Soundness of Aggregate by Use of Magnesium Sulphate.
C117	Materials Finer than 75-µm (No. 200) Sieve in Mineral Aggregates by Washing
C136	Sieve Analysis of Fine and Coarse Aggregates
D2419	Sand Equivalent Value of Soils and Fine Aggregate
D4318	Liquid Limit, Plastic Limit, and Plasticity Index of Soils
D4791	Flat Particles, Elongated Particles, or Flat and Elongated Particles in Coarse Aggregate
D6928	Resistance of Coarse Aggregate to Degradation by Abrasion in the Micro-Deval Apparatus
MoTT	**Title of Test**
APPENDIX 202-A	Fracture Count on Coarse Aggregate
APPENDIX 202-B	Petrographic Analysis Test

Table 202-B: Aggregate Properties

TEST	Test Result							
	Surfacing Aggregates	Base Course			Subbase Aggregates			Bridge End Fill
	HFSA	25mm	50mm	75mm	SGSB	IGSB	OGSB	BEF
Sand Equivalent	≥ 20	≥ 40	≥ 40	≥ 40	≥ 20	≥ 20	≥ 20	≥ 20
Micro-Deval loss factor	≤ 25%	≤ 25%	≤ 25%	≤ 17%	≤ 30%	≤ 25%	≤ 25%	≤ 30%
Fractured Faces Method "A" [1]	≥ 50[1]	≥ 50[1]	≥ 50[1]	≥ 50[1]	n/a	≥ 50[1]	≥ 50[1]	n/a
Flat and Elongated Particles using 4:1 ratio (%)	n/a	≤ 10	≤ 10	≤ 10	n/a	n/a	n/a	n/a
Plasticity	≤ 6	n/a	n/a	n/a	n/a	n/a	n/a	n/a

[1] Values are for total sample.

202.07 Source of Supply – The Contract documents may identify one or more pits or quarries, for use as aggregate sources at the Contractor's discretion.

The Ministry will provide those gravel pits or quarries and the right-of-way to those pits or quarries and will pay the cost of royalties. The cost of constructing and maintaining access roads shall be considered as part of the cost of producing the construction aggregates and shall be payable by the Contractor. The Ministry will exercise its best judgement in the selection of the pit or quarry sites; however, the failure of the pit or quarry to yield materials of uniform quality shall not constitute grounds for a claim by the Contractor against the Ministry.

The Contractor shall be satisfied as to the gradation and other characteristics of the raw material in the Ministry provided pit or quarry and as to the nature and amount of work required to produce materials which will meet all gradation requirements, and the Contractor shall, at the Contractor's expense, remove any objectionable materials from the aggregate.

202.08 Development of Pit or Quarry – Before any material for aggregate is excavated from the pit or quarry, the approved development area shall be cleared and grubbed, and all debris disposed of as ordered by the Ministry Representative. If topsoil or undesirable overburden exists, these materials shall be stripped and placed in separate stockpiles as shown on the Ministry pit development plan or as directed by the Ministry Representative.

The Contractor shall not operate the pit or quarry in a manner which will contaminate remaining granular material nor leave the pit or quarry in a condition which will limit its future use. The Contractor shall follow the Ministry pit development plan, where provided. Stockpile areas in Ministry-provided sources shall be per the pit development plan. If the Ministry pit development plan conflicts with the Health, Safety, and Reclamation Code for Mines in British Columbia, then the code shall prevail.

202.09 Use of All Materials – Any Ministry provided pit or quarry, once entered and developed, shall be used to its full potential. Normally, the Contractor shall produce select granular subbase by selection or screening, provided that oversize reject aggregate is removed and properly placed in a stockpile for later crushing and use. However, if excessive (5% or greater) oversize occurs, the Ministry Representative may order that pit run aggregate be crushed to produce 75 mm crushed granular base for use in place of select granular subbase.

For the production of crushed surfacing and base aggregates, as well as IGSB and OGSB, the Contractor shall provide crushing equipment such that all aggregate which will pass through 375 mm x 450 mm slotted openings shall be used for the production of crushed aggregate; rocks which will not pass through these openings shall be stockpiled as shown on the pit development plan or disposed of to the satisfaction of the Ministry Representative. Crushing and screening equipment shall be provided with adequate facilities and capacity to bleed off reject aggregate in usable condition, or otherwise remove any excess of fine aggregate, dust or objectionable aggregate coatings that may be present in or on the aggregate, so as to make it generally acceptable for use. No portion of the products of crushers or screening plants that can be used shall be wasted, but shall be stockpiled or used as directed by the Ministry Representative.

Table 202-C: Aggregate Gradation

Sieve Size (mm)	Percent Passing (%) Sieve Size															
	Surfacing Aggregate	Base Course Aggregates								Subbase Aggregates				Bridge End Fill
	HFSA	WGB			IGB			OGB			SGSB	IGSB	OGSB	BEF
		25mm	50mm	75mm	25mm	50mm	75mm	25mm	50mm	75mm				
75	—	—	—	100	—	—	100	—	—	100	100	100	100	100
50	—	—	100	—	—	100	55 - 100	—	100	70 - 100	—	55 - 100	70 - 100	30 - 100
37.5	—	—	80 - 100	60 - 100	—	60 - 100	40 - 80	—	75 - 100	50 - 85	—	40 - 80	50 - 85	—
25	100	100	—	—	100	40 - 75	—	100	—	—	—	—	—	—
19	85 - 100	80 - 100	50 - 100	35 - 80	65 - 100	—	17 - 40	75 - 100	35 - 65	15 - 55	15 - 100	17 - 40	15 - 55	20 - 100
12.5	—	—	—	—	—	15 - 40	—	—	—	—	—	—	—	—
9.5	60 - 85	50 - 85	35 - 75	25 - 60	30 - 70	—	—	30 - 65	5 - 35	—	0 - 100	—	—	—
6.3	—	—	—	—	—	—	—	—	—	0 - 20	—	—	0 - 20	—
4.75	40 - 70	35 - 70	25 - 55	20 - 40	15 - 40	—	—	5 - 30	0 - 15	—	—	—	—	10 - 60
2.36	—	25 - 50	20 - 40	15 - 30	10 - 30	10 - 25	10 - 25	0 - 10	0 - 10	0 - 10	—	10 - 25	0 - 10	—
1.18	20 - 40	15 - 35	15 - 30	10 - 20	—	—	—	—	—	—	—	—	—	6 - 32
0.600	—	—	—	—	—	—	—	—	—	—	0 - 100	—	—	—
0.300	10-25	5-20	5-15	3-10	5-15	5-15	4-15	0-8	0-8	0-8	0-15	4-15	0-8	4-15
0.075	7-12	0-5	0-5	0-5	0-5	0-5	0-5	0-5	0-5	0-5	0-5	0-5	0-5	0-5

202.10 Modification of Materials – The rejection or addition of any particular size material necessary to produce aggregate conforming to specification requirements shall be the responsibility of the Contractor, and no additional compensation will be paid therefore.

202.11 Change of Ministry Provided Pit or Quarry – No payment will be made to the Contractor for moving the plant from one Ministry provided pit or quarry to another Ministry provided pit or quarry.

However, if the Ministry Representative requires a move to a new Ministry provided pit or quarry, the Contractor will be given notice in writing and upon completion of the move, the Contractor will be paid a lump sum of $7,500 for moving the existing crushing operation and any additional equipment that may be required for material processing at the new pit. The payment will be deemed to be full compensation for all costs and expenses of the move and set-up of the crusher. If the Contractor moves the crushing plant without first securing a written order to do so from the Ministry Representative it will be considered sufficient proof that the move was unnecessary, and no allowance or compensation will be made to cover such a move.

202.12 202.12 Pit Cleanup – When the Contractor discontinues operations in any pit or quarry, the Contractor shall trim the face of pit excavations, waste piles and stockpiles to a 1.5H to 1V slope, or such other slope as the Ministry Representative may direct, open up such drains or ditches as may be required to prevent water standing therein, and leave the site in a neat condition, all to the satisfaction of the Ministry Representative. The sloping of pit excavations shall be with in-place materials and shall not be contaminated with topsoil, overburden or any reject aggregates. The trimming, sloping, ditching and draining of the pit or quarry, as well as maintenance of the access roads, shall be considered part of the cost of producing the granular aggregate and no additional payment will be allowed.

If a pit has been fully depleted and is to be abandoned, the Contractor may be requested to reshape the pit faces to a slope not steeper than 2H to 1V using the stockpiled overburden if necessary, and cover same with any topsoil previously removed and stockpiled, all as directed by the Ministry Representative. In such cases, the replacing of stockpiled overburden and topsoil will be paid for in accordance with SS 202.32.

202.13 202.13 Work in Ministry Pits or Quarries – All work in a Ministry pit or quarry shall be in accordance with SS 145.26.

CONTRACTOR SUPPLIED AGGREGATES

202.14 Preliminary and General – Aggregates shall meet the requirements of SS 202.02 to SS 202.05 inclusive.

202.15 Development and Clean Up of Private Pits or Quarries – Pit or quarry development and cleanup shall be in accordance with local and Provincial regulatory agency requirements.

202.16 Contractor to Pay All Costs – The Contractor shall bear and pay all costs including, applicable royalties, construction and maintenance of access roads to borrow sites, mobilization/demobilization of weigh scales, and any move of the equipment that may be ordered or required because a pit or quarry of the Contractor's choosing proves unsuitable for production of the contracted quantity of granular aggregates meeting the requirements of this SS 202.

PRODUCTION OF CRUSHED AGGREGATE

202.17 Equipment and Plant – The Contractor shall provide and maintain in good operating condition, all plant and equipment necessary to comply with the requirements of this specification. Such plant and equipment shall be of a type and capacity to ensure proper construction and maintenance of access roads, development of the pit or quarry, maximum use of the pit or quarry, production of granular aggregates to the standard of uniformity required, and adherence to the construction schedule.

Adequate sampling facilities shall be installed by the Contractor at the discharge end of the production conveyor.

The Contractor shall ensure that the plant and equipment meet the requirements of SS 145.

202.18 Working of Pit – A pit shall be worked in such a manner as to ensure that granular material excavated and fed to the crushers, screening plants or loaders shall be as uniform as possible. This may require the use of equipment capable of deep excavating through horizontal gravel layers or carrying out selective digging or cross blending over a large area of the pit. Before commencing production, the Contractor's quality control plan shall describe the proposed method of production for the Ministry Representative and obtain the Ministry Representative's general acceptance.

202.19 Crushing Operations – The Contractor shall adjust the crushers and/or screening plants and provide such screens as may be necessary to produce and maintain acceptable gradations of granular aggregates.

202.20 Crusher Gradation Control

202.20.01 Design Aggregate Gradation – For each aggregate gradation classification specified for production, field and/or laboratory tests shall be performed by the Contractor to select a single gradation curve within the "banana" specification (Table 202-C), the "Design Aggregate Gradation" (DAG), which will be the target for crusher control. The Contractor shall declare the DAG, in writing to the Quality Manager and the Ministry Representative, within production of the first 10% of the Approximate Quantity of a given aggregate classification.

As part of the Ministry's quality assurance program and for payment purposes, the Ministry may sample the stockpile to assess aggregate gradation compliance to contractual requirements.

202.20.02 Variation Limits - The running average of four (4) consecutive tests, on each applicable individual sieve size specified in Table 202-C, must be maintained at all times within the limits specified in that Table. Also, each individual test must be within the specifications of Table 202-C.

The maximum permissible variation from the DAG curve, on each applicable individual sieve size specified in Table 202-C of the mean of any four (4) consecutive tests shall also be within the limits specified in Table 202-D.

202.20.03 Field Adjustment of Design Aggregate Gradation - The Contractor may make a maximum of two (2) field adjustments to the Design Aggregate Gradation, to accommodate minor changes in pit characteristics or to enhance production efficiency.

A field adjustment to the Design Aggregate Gradation is defined as a change in declared target gradation of the various aggregate sizes within the specified limits shown in Table 202-C which does not require a supplementary formal review. The maximum adjustment for individual sieve sizes for each adjustment are as permitted in the Table 202-E.

A proposed Design Aggregate Gradation field adjustment shall be submitted in writing with supporting documentation (showing original DAG, current stockpile average gradation, and proposed field adjustment and the new DAG) to the Ministry Representative. Upon receipt of the proposed field adjustments, the Ministry Representative will assess the adjustment for conformance with the contract requirements and notify the Contractor whether or not it is acceptable in a timely fashion.

Table 202-D: Aggregate Gradation Variation Limits

Sieve Size (mm)	Variation Limits (% Passing)
4.75and larger	± 5
1.18 to 2.36	± 3.5
0.300 and 0.600	± 2
0.075	± 1

Table 202-E: Field Adjustment of Design Aggregate Gradation

Sieve Size (mm)	Maximum Field Adjustment (% Passing)
19 and larger	± 3.0
9.5	± 2.5
2.36 and 4.75	± 2.0
0.600 and 1.18	± 1.5
0.150 and 0.300	± 1.5
0.075	± 1.0

No field adjustment will be acceptable if it results in a change from the current Design Aggregate Gradation to fall outside the limits defined in Table 202-C. The variation limits specified in SS 202.20.02 will apply to the adjusted DAG.

The Contractor shall have the moisture/density relationship established per ASTM D698 for Standard Proctor Maximum Dry Density for each original DAG, and one after each field adjustment thereto.

202.20.04 Declaration of a New Design Aggregate Gradation (DAG) - Should there be a substantial change in the character of aggregate exposed in the pit face as the work proceeds, the Ministry Representative may authorize a change in the declared Design Aggregate Gradation, which would then also be eligible for field adjustment in accordance with SS 202.20.

A maximum of one change to the declared DAG will be permitted.

202.21 Stockpiling

202.21.01 Stockpile Before Use – Unless it is specified in the Special Provisions or ordered otherwise in writing by the Ministry Representative, all crushed aggregates shall be stockpiled prior to use on the Highway.

202.21.02 Site Preparation – Stockpile sites shall be cleared of all vegetation, trees, brush, rocks or other debris and a uniform gravel surface prepared before the stockpile material is deposited on the stockpile site.

202.21.03 Stockpile Shape – Stockpiles shall be constructed on the designated site and when completed shall be neat and regular in shape, occupying as small an area as is practicable. Spilling of material over the edges of the piles will not be permitted.

202.21.04 Layer Thickness – The Contractor shall ensure that stockpiles shall be built up in layers not to exceed 1 m in thickness.

202.21.05 Protection Measures – The Contractor shall ensure that plank or protected runways shall be provided for operating trucks on stockpiles when the Ministry Representative deems them necessary to prevent dirt being tracked onto the crushed aggregate.

202.21.06 Bulldozing Aggregates – The Ministry Representative may, on receipt of a written request from the Contractor, permit the Contractor to build the final stockpile by bulldozing the aggregate from a feed pile at the end of a production belt provided that the following are maintained:

(a) The bulldozer(s) to be used shall be equipped with U-shaped pushing blades.

(b) The aggregate does not become contaminated with oversize material, mud or other objectionable material picked up from the pit floor or general working areas.

(c) The crushed granular aggregate is distributed evenly over the final stockpile area in lifts not greater than 150 mm in depth.

(d) The cone of the feeder pile at the end of the production belt shall not, without express permission, be allowed to build up to a height greater than 2 m.

(e) No appreciable segregation or degradation shall occur in the main stockpile as a result of using this method.

202.21.07 Radial Stackers – Radial stackers may be utilized only with the approval of the Ministry Representative and under Contractor-developed written work and quality management procedures governing the stockpiling operation that ensures that:

(a) the stockpile is large enough to accommodate all required material, or that separate stockpiles are developed;

(b) segregation does not occur; and

(c) lift thicknesses, drop heights, and wind-driven fines migration are minimized.

CONSTRUCTION METHOD

202.22 Target Density for Compaction – For each material produced and placed under this SS 202, the Contractor shall determine a Target Density in accordance with the following.

(a) Laboratory Density – Where the material gradation is within the applicable range, the Target Density shall be the Standard Proctor Maximum Dry Density obtained under the current ASTM D698, with oversize correction in accordance with ASTM D4718; or

(b) Field Density – Where the material falls outside the applicable range of ASTM D698 and ASTM D4718, the Target Density shall be 98% of the maximum dry density achieved through field compaction at varying moisture content and varying number of compaction passes. The methodology employed in determining the field density shall be developed by the Contractor and its Quality Manager, and be subject to the approval of the Ministry Representative

If the Contractor's methodology is not approved by the Ministry Representative, then the Contractor may propose an alternative method or the Ministry Representative may order that each lift or course of aggregate shall be continuously and thoroughly rolled until successive passes of a vibratory roller results in an increase in density of less than 10 kg/m3.

The vibrating roller shall have a minimum steel drum diameter of 1.15 m, a minimum drum width of 1.5 m, and shall be capable of being loaded so as to have a gross mass of 20 kg per lineal centimeter of drum width.

However, if the Contractor elects to use alternate compaction equipment, then upon written request by the Contractor, the Ministry Representative may give written permission for the substitution of new or alternative compaction equipment if the Ministry Representative is satisfied that such equipment will provide equal or superior compaction performance.

(c) Proof Rolling – The Ministry Representative may require a proof rolling, in accordance with SS 202.29, at any time to assess the performance of the grade at the Target Density.

202.23 Bridge End Fill – Construction of the bridge end fill shall not commence until the Ministry Representative has assessed the pertinent foundation conditions that may affect the future stability of both the bridge and embankments and has authorized continuation of construction. Unless the procedure for construction is stated in the Special Provisions, the Contractor's proposed method must be submitted to the Ministry Representative for approval before commencing work.

Bridge end fill shall be constructed to the specified thickness and dimensions as shown on the Drawings or as described in the Special Provisions, unless otherwise directed by the Ministry Representative. The bridge end fill shall be constructed to the subgrade elevation.

The material shall consist of mineral soil with properties and gradation in accordance with SS 202.04 and SS 202.05.

The bridge end fill shall be constructed in successive horizontal layers not exceeding 150 mm in loose thickness, compacted to meet or exceed the applicable Target Density determined in accordance with SS 202.22.

202.24 Thickness of Granular Courses – Crushed surfacing course, crushed base courses and subbase courses shall be constructed to the specified thickness and dimensions as shown on the Drawings or as described in the Special Provisions, unless otherwise directed by the Ministry Representative.

Normally, on new construction, crushed bases shall be constructed 300 mm thick (in individual lifts with compacted thickness of 150 mm) unless otherwise approved by the Ministry Representative. Additional base or subbase granular materials will be of such thickness as may be considered necessary to provide supporting strength for the flexible pavement structure.

202.25 Construction of Subbase

202.25.01 Spreading – Aggregates for subbase shall be delivered to the roadbed as uniform mixtures and shall be spread in layers without segregation, preferably through an approved aggregate spreader. Granular aggregate shall not be end-dumped from trucks in piles on the grade. The Ministry Representative may permit spreading from the tailgate of trucks or from centre dump units, provided the

Ministry Representative is satisfied that the work will be well controlled and segregation will not occur.

When the subgrade below granular aggregate subbases consists of material susceptible to damage from wheeled equipment, and written permission is granted by the Ministry Representative, a portion of the granular aggregate subbase or base may be dumped in piles upon the subgrade and spread ahead in sufficient quantity to stabilize the subgrade. Segregation of aggregates shall be avoided and the material as spread shall be free from pockets of large or fine material. Segregated materials shall be remixed until uniform. Static compaction shall be used in areas of sensitive soil subgrades.

Where the required thickness is 150 mm or less, the granular base or subbase may be spread and compacted in one layer. Where the required thickness is more than 150 mm, the aggregate shall be spread and compacted in two or more layers of approximately equal thickness; the maximum compacted thickness of any one layer shall not exceed 150 mm. Each layer shall be spread and compacted in a similar manner. At locations where the granular base or subbase is to be placed over areas susceptible to damage by the spreading equipment, the granular subbase or base may be spread by any means acceptable to the Quality Manager and the Ministry Representative to obtain the specified results.

202.25.02 Compaction – Immediately following spreading, the material shall be compacted to meet or exceed the applicable proportion of the Target Density.

202.25.03 Watering – The Contractor may water the subbase aggregates as required to aid in attaining the specified density, but shall not ensure that overwatering does not occur.

202.25.04 Completed Surface Accuracy – The completed surface of the subbase course shall conform to the required line, grade and cross section as shown on the Drawings to an accuracy of ± 15 mm, neither uniformly high nor low.

202.26 Construction of Crushed Base Course

202.26.01 General – Crushed base course aggregate shall be placed on a properly prepared subbase surface to such depth or at such rates as may be specified.

If the Ministry Representative is of the opinion that the finished surface of the subbase does not meet the requirements of SS 202.25.04 but has been thoroughly and densely compacted and should not be disturbed, the Ministry Representative may order that the surface of the subbase be corrected to the true cross section, line and grade, and within the tolerances specified by use of a levelling course of crushed 25 mm base course aggregate. In such event, the crushed 25 mm base course aggregate so used will be paid for only at the rates bid for subbase, provided the same Contractor is responsible for the construction of both the crushed base, and subbase.

The crushed base course shall be constructed in such a manner that the aggregate is neither segregated, contaminated nor degraded. End dumping will not be permitted. The thickness of the crushed base course shall be substantially uniform and the minimum thickness shall not be less than the nominal thickness shown on the Drawings or ordered by the Ministry Representative. If the Contractor is unable to provide adequate manually operated equipment or workers of sufficient skill to lay the crushed base course aggregate within the tolerances specified, the Ministry Representative may require that the Contractor lay the aggregate through an approved electronically controlled spreading machine.

202.26.02 Compaction – Immediately following spreading, the crushed base course aggregate shall be compacted to meet or exceed the applicable Target Density.

The method of compaction to be employed may be selected by the Contractor, but shall be subject to approval by the Quality Manager and the Ministry Representative

202.26.03 Watering – Subject to the agreement of the Ministry Representative, the Contractor may water the base course aggregates as required to aid in attaining the specified density.

202.26.04 Tolerance of Completed Surface – The completed surface of the granular base course shall conform to the required line, grade and cross section as shown on the Drawings to an accuracy of ± 10 mm, neither uniformly high nor low.

202.27 Equipment for Watering – Water shall be applied from a pressure type distributor, equipped with spray bar mounting nozzles similar to those used on asphalt distributors and capable of applying the water accurately and uniformly. Splash plate type distributors or those equipped with spray bars that eject fine streams of water will not be permitted. The distributor must be provided with a satisfactory means for accurately measuring the quantity of water sprayed. If the Ministry Representative so requires, measuring equipment shall be calibrated under the Ministry Representative's inspection. All necessary arrangements and permits for obtaining water shall be performed by the Contractor and at the Contractor's expense.

202.28 Weather and Job Conditions

202.28.01 General – No construction shall be undertaken during snow, heavy rain, freezing or other unsuitable conditions. Aggregate shall not be placed upon a frozen, wet, muddy, rutted or undulating or rutted subgrade, subbase, base or surface unless otherwise directed by the Ministry Representative.

202.28.02 Excessive Moisture – When the subgrade is soft due to excessive moisture, the placing of granular subbase shall be stopped until rutting or displacement of the initial subbase layers can be prevented by reduced loading or by other means approved by the Ministry Representative.